

Distress Lexicon — Implementation Report

LendingClub Default-Prediction Thesis — Level 1 (Lexicon) of the 3-Tier Text-Representation Framework

Protocol source: *thesis_lexicon_plan_updated_260820_023907.pdf*. Generated by `Dictionaries/scripts/06_generate_phase1_report.py` from the pipeline's own audit artifacts (never hand-typed).

Overview

This report documents the Distress Lexicon build end-to-end: (1) frozen vocabulary extraction from the initial walk-forward training window, (2) zero-shot classification of that vocabulary by Claude via the real Anthropic API, (3) blind human-expert validation of a stratified sample against the LLM's own labels, (4) application of the frozen lexicon to the full loan corpus to produce per-loan features, and (5) the downstream walk-forward predictive comparison (Structured vs Structured+Lexicon) across all 14 folds.

Step 1 — Strict Vocabulary Extraction & Freezing

Per the protocol's no-data-leakage requirement, the vocabulary was extracted **exclusively** from the initial baseline training window — the same walk-forward fold-boundary definition already used by `TF-IDF/tfidf_pipeline.py` and `analysis/preliminary_results_v2.py` (`MIN_TRAIN_MONTHS=8`, date column `issue_month_start`).

Parameter	Value
Initial window	2010-01 .. 2010-08
Rows in window (is_default in [0,1])	6,694 / 217,287 total rows
Non-empty desc rows in window	4,503
Document-frequency pruning	16 (i.e. freq > 15, per the PDF)
N-gram range / top-N cap	(1, 2) top-N cap: none (uncapped)
Vocabulary size	1,650 (Unigrams: 1,015, Bigrams: 635)

The Prompt Sent to Claude

Stored verbatim in `Dictionaries/prompts/zero_shot_lexicon_rubric_v1.json` (version **v1**), referenced by the annotation script rather than re-typed — the exact text sent on every call is inspectable and citable.

System prompt (verbatim from the protocol PDF, section 3):

You are an expert quantitative credit risk analyst. Your objective is to annotate a vocabulary list extracted from peer-to-peer personal loan descriptions. Classify each term into EXACTLY

ONE of the following categories: - DEBT_PRESSURE: Terms signaling high debt load, maxed credit lines, mounting interest, or debt-spiral traps. - LIQUIDITY_SHORTAGE: Terms signaling immediate cash-flow deficit, inability to cover basic living expenses, or being past due. - SHOCK_EMERGENCY: Terms signaling unexpected exogenous financial distress (medical emergencies, sudden job loss, urgent repairs). - HIGH_RISK_REFINANCING: Terms explicitly indicating payoff of predatory/payday debt or high-interest short-term borrowing. - NEUTRAL_BENIGN: Terms describing routine loan logistics, stable employment, positive sentiment, planned life events, or general conversational prose. Output format (CSV): token,category,is_distress(0/1),rationale

User-turn template (this project's own addition — the PDF specifies only the system rubric):

Classify each of the following {n} terms per the rubric above. Output only the CSV rows, one per term, in the same order given, no header, no commentary, no markdown code fences. Rationale must be a short phrase (under 12 words) with no commas; if you must use a comma, wrap the whole rationale field in double quotes per standard CSV rules: {terms}

Step 2 — Zero-Shot LLM Annotation

Parameter	Value
Model	`claude-opus-5`
anthropic package version	`0.125.0`
output_config.effort	low
Chunk size / chunks sent	50 terms/request Chunks sent: 33
Terms classified	1,649 / 1,650 requested
Wall-clock time	523.2s

Known deviation from the protocol: the PDF specifies "Temperature fixed at 0.0 (strict determinism)". Claude Opus 5 no longer accepts a temperature parameter at all (HTTP 400 if sent) — there is no substitute knob for byte-identical determinism on this model generation. No temperature parameter was sent; effort=low with adaptive thinking is the closest available approximation. This should be documented as a methodological footnote in the thesis text wherever "T=0.0" is claimed.

Category distribution (1,649 terms classified)

Category	Count	% of classified terms
NEUTRAL_BENIGN	1337	81.1%
DEBT_PRESSURE	199	12.1%
LIQUIDITY_SHORTAGE	66	4.0%
SHOCK_EMERGENCY	40	2.4%
HIGH_RISK_REFINANCING	7	0.4%
Total distress (excl. NEUTRAL_BENIGN)	312	18.9%

Step 3 — Human-in-the-Loop Validation

A stratified random sample of **147 terms** was drawn across the 5 categories and blind-labeled by Ariel (human expert), without visibility into the LLM's own labels for the same terms. The protocol targets a 150–200 term sample; HIGH_RISK_REFINANCING has only 7 terms in the full lexicon, which capped that category's contribution and brought the achieved sample to 147 rather than the full 150+ target.

Agreement metrics

Metric	Value	Interpretation
Cohen's kappa (5-category)	0.556	Moderate agreement (Landis and Koch scale)
Raw agreement (5-category)	66.0%	—
Binary is_distress precision	0.688	~31% of LLM-flagged terms are, per the human rater, not distress
Binary is_distress recall	0.975	LLM rarely misses a term the human considers distress
Binary is_distress F1	0.806	—

Per-category precision / recall (human label = ground truth)

Category	Precision	Recall	F1
DEBT_PRESSURE	0.257	0.562	0.353
LIQUIDITY_SHORTAGE	0.743	0.788	0.765
SHOCK_EMERGENCY	0.771	1.000	0.871
HIGH_RISK_REFINANCING	0.286	0.667	0.400
NEUTRAL_BENIGN	0.943	0.485	0.641

Key finding: the LLM over-classifies neutral terms as distress, particularly under DEBT_PRESSURE (precision 0.257 — most terms the LLM assigns to this category, the human rater considers benign). Of 68 human-labeled NEUTRAL_BENIGN terms, the LLM agreed on only 33 (48.5% recall), misclassifying 23 as DEBT_PRESSURE alone. This moderate-kappa result (0.556) should be reported as-is in the thesis as the construct-validity finding for this lexicon — it is evidence the categorical boundaries (especially DEBT_PRESSURE vs NEUTRAL_BENIGN) are less crisp for the LLM than the rubric intends, not a pipeline defect.

Confusion matrix (rows = human label, columns = LLM label)

human \ LLM	DEBT	LIQUIDITY	SHOCK	REFI	NEUTRAL
DEBT	9	2	3	2	0
LIQUIDITY	2	26	2	1	2
SHOCK	0	0	27	0	0
REFI	1	0	0	2	0

NEUTRAL	23	7	3	2	33
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Step 4 — Feature Engineering on the Full Corpus

The frozen lexicon (built from the initial window only) was applied unchanged to the full loan corpus — all folds, not just the training window — since freezing applies to vocabulary *extraction* only, not feature *application*.

Metric	Value
Total loans	217,287
Loans with >=1 distress term (risk_word_flag=1)	85,705 (39.44%)
Output file	Dictionaries/output/lexicon_features.csv

Feature columns produced (per loan, joined by id):

- risk_word_count — total distress terms matched
- risk_word_density — risk_word_count / total word count
- risk_word_flag — 1 if risk_word_count ≥ 1
- distress_debt_pressure_count
- distress_liquidity_shortage_count
- distress_medical_life_shock_count
- distress_refinancing_burden_count

Step 5 — Downstream Walk-Forward Evaluation

Structured vs Structured+Lexicon, evaluated with the same walk-forward protocol, fixed hyperparameters, and metrics as TF-IDF/tfidf_structured_ablation.py and FinBERT/finbert_structured_ablation.py: Logistic Regression, XGBoost, and Random Forest, across all 14 folds, with 95% bootstrap CIs and the DeLong (1988) pairwise significance test.

Mean AUC by arm and model ($\pm$ std across 14 folds)

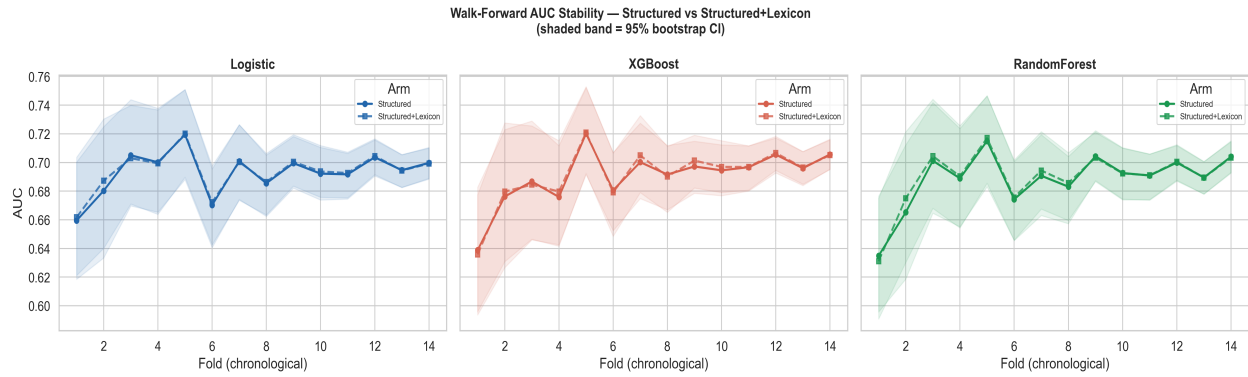
Arm	Model	Mean AUC	Std AUC
Structured	Logistic	0.6930	0.0152
Structured	RandomForest	0.6882	0.0199
Structured	XGBoost	0.6904	0.0191
Structured+Lexicon	Logistic	0.6939	0.0142
Structured+Lexicon	RandomForest	0.6895	0.0203
Structured+Lexicon	XGBoost	0.6913	0.0200

DeLong pairwise test (Structured+Lexicon vs Structured, mean across folds)

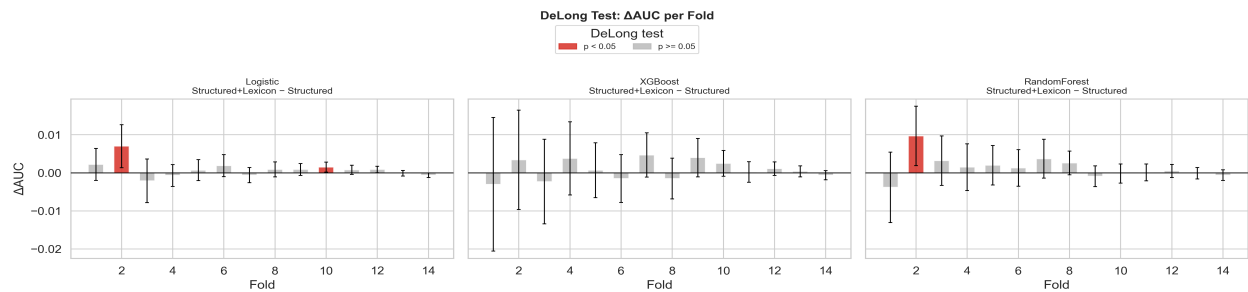
Model	Mean Delta AUC	Mean p-value	Significant folds ($p < 0.05$)
Logistic	+0.0009	0.317	2 / 14
RandomForest	+0.0014	0.494	1 / 14
XGBoost	+0.0009	0.496	0 / 14

Key finding: across all three models, adding the 7 frozen Distress Lexicon features to the structured baseline changes mean AUC by at most 0.0014 (essentially negligible), and the DeLong test finds this difference statistically significant ($p < 0.05$) in at most 2 of 14 folds per model. Consistent with the moderate Cohen's kappa (0.556) and the LLM's tendency to over-flag neutral terms (Step 3), **this frozen zero-shot lexicon, in its current form, does not add meaningful incremental predictive value over the structured features already in the model.** This is a legitimate, reportable null result for the thesis's Level 1 vs Level 2/3 comparison — not a pipeline failure.

AUC stability by fold



DeLong Δ AUC by fold



Full per-fold tables: [results/lexicon_ablation_folds.csv](#),
[results/lexicon_ablation_delong.csv](#),
[results/lexicon_ablation_feature_importance.csv](#). Full narrative:
[results/lexicon_ablation_report.md](#).

Known Limitations & Deviations

- Temperature could not be fixed at 0.0 — parameter removed on Claude Opus 5 (see Step 2).
- Human validation sample (147 terms) fell short of the 150–200 target due to small category sizes (HIGH_RISK_REFINANCING has only 7 terms total).
- One term ("return") was dropped mid-response in chunk 5 of the annotation run and was not classified — not yet manually resolved.
- Cohen's kappa of 0.556 (moderate) indicates the LLM's category boundaries, especially DEBT_PRESSURE, diverge non-trivially from the human rater's judgment — a construct-validity caveat that should be reported, not hidden.